**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 126 · Lesson 126 · Addressing Modes: Immediate, Direct, Indirect and Indexed

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Which LMC instruction reads a mailbox value into the accumulator?

Q2. In a BST, which side holds a smaller value?

Q3. How many bits are in three bytes?

Q4. What is a function parameter?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

immediate — Operand is a literal value.

direct — Operand gives the memory location holding a value.

indirect — Operand points to a location holding the effective address.

indexed — Base address plus index gives effective address.

## New learning

Immediate versus direct — LOAD #12 uses literal 12; LOAD 12 uses contents of memory[12].

Indirect addressing — LOAD @12 uses address stored in memory[12]; then loads data there.

Indexed addressing — LOAD 20,X uses effective address 20+X; useful for array elements.

## Worked example

With memory[12]=40 and memory[40]=7, compare LOAD #12, LOAD 12 and LOAD @12.

1. #12 gives literal 12 — Predict the first step from the prompt.

2. 12 reads memory[12]=40 — Explain the intermediate result or decision.

3. @12 follows pointer to 7 — Using illustrative assembler conventions: immediate gives 12; direct reads 40; indirect follows memory[12]=40 then loads memory[40]=7.

# We Do · Practise with guidance

Array starts at memory address 80. X=3 and memory[83]=19. What does LOAD 80,X load?

Base 80, index 3

Predict the first step from the prompt.

Effective address 83

Explain the intermediate result or decision.

Read memory[83]=19

Effective address = 80+3=83. It loads 19. Indexed addressing avoids writing a new absolute address for each element.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Define immediate and direct addressing with one example each. [6]

### Question 2

Memory[15]=29 and memory[29]=6. Give values loaded by direct LOAD 15 and indirect LOAD @15. [6]

### Question 3

Explain indexed addressing for an array at base 50 with index 4 and discuss why this is not simply LMC LDA notation. [8]

## Exit Ticket · Answer before checking

In direct addressing, the operand always means the literal value to be used.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. LDA (load).

Q2. The left subtree.

Q3. 24.

Q4. An input used by a subroutine.

### You Do pages · Independent questions

Q1. Immediate uses supplied literal [2]; direct accesses memory at given address [2]; valid contrasting examples [2].

Q2. Direct =29 [3]. Indirect follows pointer 29, then loads 6 [3].

Q3. Effective address base + offset =54 [3], retrieves element stored there [2], supports array traversal [1], standard LMC has a restricted instruction set and does not natively specify general indexed addressing [2].

### Exit Ticket · Model answer

False. Direct uses the operand as a memory address; immediate uses a literal.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**